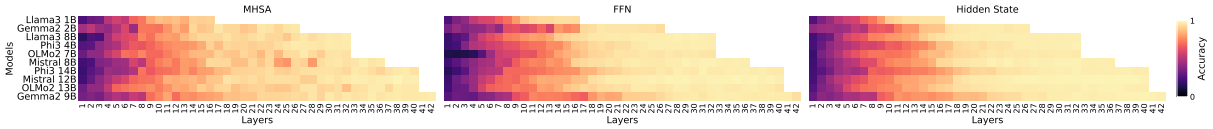

Figure 2: Layer-wise accuracies of emotion probe experiments across different models (each row) with varying depths at (**Left**) MHSA, (**Mid**) FFN, and (**Right**) hidden states. The results suggest an increasing signal with clear consolidation in the mid layers across various model families and sizes, which indicates that models predominantly make emotion-related decisions by the mid layers with minimal improvement in higher layers.

$$\mathbf{h}_t^{(l)} = \mathbf{h}_t^{(l-1)} + \mathbf{a}_t^{(l)} + \mathbf{m}_t^{(l)},$$

where $\mathbf{a}_t^{(l)} \in \mathbb{R}^d$ and $\mathbf{m}_t^{(l)} \in \mathbb{R}^d$ are MHSA and FFN outputs at layer l for token t. $\mathbf{h}_{1:t}^{(l-1)}$ are the previous layer's hidden states for tokens 1 to t.

Throughout this paper, we focus on activations $\mathbf{x}_t^{(l)}$ selected from one of the three candidates in $\{\mathbf{a}_t^{(l)}, \mathbf{m}_t^{(l)}, \mathbf{h}_t^{(l)}\}$ and study their properties at different layers and token positions. While activations can be extracted from any token index, we anticipate the strongest emotion signals to be present at the last token, as it directly influences the model's output prediction in our task setup. Therefore, when clear from context, we omit the subscript T while studying the last token. Also, we drop the superscript (l) when generally discussing any activation across different layers.

5 Probing for Emotion Signals

Building on the linear representation hypothesis (Mikolov et al., 2013b; Elhage et al., 2022; Park et al., 2024), we perform probing experiments to assess the presence and strength of emotion-related signals at different activations within the model. Specifically, we train linear classifiers (Hewitt and Liang, 2019) to predict the corresponding emotions. We formalize the linear classifiers as follows:

$$\hat{\mathbf{y}} = \mathbf{W}^\top \mathbf{x} + \mathbf{b},$$

where $\mathbf{x} \in \mathbb{R}^d$ denotes the activation vectors at one of the locations specified in the previous section. $\mathbf{W} \in \mathbb{R}^{d \times C}$ is the weight matrix for emotion classification, $\mathbf{b} \in \mathbb{R}^C$ is the bias vector, C represents the number of emotion classes, and $\hat{\mathbf{y}} \in \mathbb{R}^C$ denotes the predicted logits for each emotion class.

We perform probing separately over different activation locations and layers across the model for the last token index. The probing results in Figure 2, measured as the accuracy on a held-out test set, indicate that the models begin consolidating emotional information in the hidden states $\mathbf{h}^{(l)}$ neither too early nor too late, but predominantly around the mid-layers across all models. For example, in the first row corresponding to Llama 3.2 1B in Figure 2, the emotional signal peaks by layer $l = 10$ out of a total of 16 layers. Beyond layer 10, there is no significant increase in probe accuracy, suggesting that the model effectively captures emotional content by this stage.

There is no clear distinction in probing performance between $\mathbf{m}^{(l)}$ and $\mathbf{h}^{(l)}$. Measurements from FFN closely track the hidden state dynamics, showing a steady increase in emotional conceptualization that peaks around the mid-layers. However, the heatmap corresponding to $\mathbf{a}^{(l)}$ shows a more dispersed pattern while following the same consistent increasing trend observed in other locations.

Our experiments reveal that emotion-processing mechanisms in LLMs are most pronounced in the middle layers across model families and sizes. *These findings suggest that the model has largely determined the output emotion by the mid-layers, with subsequent layers adding little additional processing.* Our observation aligns with the understanding that higher transformer layers capture more abstract and task-specific features.

Lastly, to evaluate the hypothesis regarding the importance of the last token in causal modeling, we repeat the analysis on the last five tokens for Llama 3.2 1B in Appendix C.3. We observe a consistent increase in signal strength from earlier to later tokens, reinforcing the focus on the last token as the primary contributor to output generation.

6 Emotion Transfer by Activation Patching

Given evidence suggesting that the model's internal representation of emotional content stabilizes around the mid-layers, we explore causal intervention in these regions to test their functional importance. Specifically, we assess whether the output emotion of a *source* example can be transferred to a *target* example, with a different emotion, by selectively patching activations from the source com-

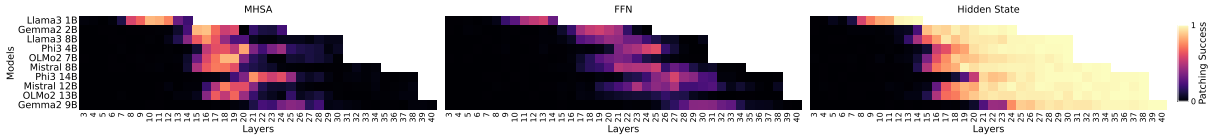

Figure 3: Results of activation patching experiments where we measure the success rate of transferring the output emotion by patching an activation from a source sample to a target sample. The patching is performed at (**Left**) MHSA, (**Mid**) FFN, and (**Right**) hidden states, respectively. The MHSA and FFN heatmaps demonstrate a clear localization with successful transfer peaks happening in the mid-layers consistently across various model families and scales. This observation aligns with the consolidation points observed in the probe heatmap (Figure 2) and indicates how activation patching can identify critical regions for emotion prediction.

putation graph into the inference pass of the target at corresponding locations, a method referred to as activation patching (Ghandeharioun et al., 2024).

Formally, let $\mathbf{x}_t^{(l)}$ be the activation vector from the target example, and $\hat{\mathbf{x}}_t^{(l)}$ be the activations from a different example, i.e. the source sentence, which has a mismatched label from the target. The patching operation involves replacing the activation at layer l and token t by substituting $\mathbf{x}_t^{(l)} \leftarrow \hat{\mathbf{x}}_t^{(l)}$ and letting the model continue the processing flow in the following layers and tokens.

The goal is to determine whether substituting specific activations with those from another example can manipulate the model's final prediction to reflect the intended label. We conduct experiments by substituting activations at the last token and within a window spanning five layers, consistently across all model families and sizes[1].

The results shown in Figure 3 demonstrate consistent behavior across model sizes and families. When intervening on hidden state activations, we observe a clear increase in intervention success (see Figure 3 right). Take Llama 3.2 1B as an example. Patching hidden states in the early layers is entirely ineffective. There is a critical point, e.g. layer 10 in this model, after which copying the hidden state transfers the emotion label with a high success rate. A high success rate at the final layers is naturally an expected behavior, as the residual stream of hidden states aggregates the information as it gets closer to the final layers. However, remarkably, the chance of success peaks and stabilizes around the mid-layers in all models, an observation that aligns with our findings in the previous section.

To dig deeper, we look at the patching effect of MHSA and FFN units (see Figure 3 left and middle, respectively), which provides clear evidence for functional localization of emotion processing. More precisely, interventions targeting $\mathbf{a}^{(l)}$ and $\mathbf{m}^{(l)}$ show clear evidence of success localized to specific mid-layers, e.g., MHSA units of layers $l \in [9-11]$ in Llama 3.2 1B. In other words, successful patching of both MHSA and FFN units occurs only in a subset of layers and predominantly in the middle rather than the final layers, with the FFN's most successful patchings happening only slightly later than the MHSA's patching. *We hypothesize that there are a few consecutive layers in each language model whose MHSA units are responsible for gathering emotional information from the rest of the tokens and integrating it into the hidden state of the last token. This mechanism is immediately followed by a processing in the subsequent FFN units.*

To complement these findings, we perform an additional experiment, where a set of activations is knocked out in the forward pass to assess their impact. The results in Appendix B consistently align with those of our activation patching and probing, reinforcing the evidence of functional localization in emotion processing. These observations hold across different models and experimental methods.

Our results in all previous sections are not prompt-dependent. Specifically, changes in the format, wording, structure, or the number of demonstrations in the prompt do not affect the findings (see Appendix C.5). Additionally, we provide a control experiment in Appendix C.4 to show that the identified units are not critical when performing a different yet syntactically similar task. In fact, the final layers are most critical for this isomorphic syntactic task, which differs significantly from the units we found for emotion processing.

To further explain our findings, we analyze the attention patterns in Llama 3.2 1B. Specifically, we record the top 3 tokens attended to by all attention heads in the last token of each layer. We conduct this analysis for all samples in the dataset, provid-

[1] Smaller models are more sensitive to fewer transferred layers, while larger models resist emotion transfer and require a larger span. For consistency, we use the same window size across all models.

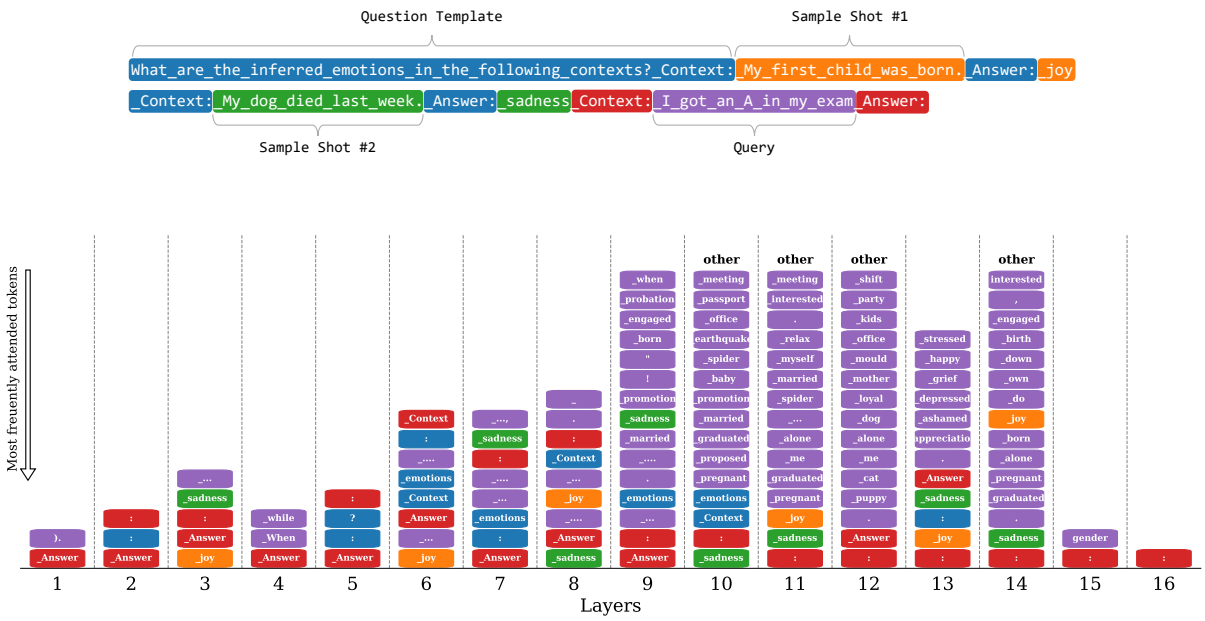

Figure 4: (**Top**) Primary prompt used in this study. Different segments of the prompt are colored differently. (**Bottom**) Most attended tokens at each layer from the perspective of the last token in Llama 3.2 1B. Layer 9 is the first layer in which the model attends to the tokens in the query with high emotional importance.

ing insight into which tokens are most frequently attended to. Figure 4 presents the aggregate attention patterns. The results indicate that early layers primarily focus on syntax. Around the mid-layers, the model shifts its attention to emotionally relevant tokens in the query and maintains this focus until the final layers. In the last few layers, attention predominantly focuses on the last token in the sequence, suggesting that it primarily carries forward the last token's hidden state from earlier layers. These patterns provide evidence that the identified middle units are meaningfully related to emotion processing.

7 Investigating Appraisal Concepts

We draw inspiration from cognitive appraisal theory to show an existing emotion structure in LLMs' latent representations. Appraisals are known to have significant associations with emotions and, in some accounts, are considered causal factors in their emergence (Rosenman and Smith, 2001).

Figure 5 illustrates three primary appraisal dimensions, including *pleasantness*, *other-agency*, and *predictability*, in addition to their associations with a set of basic emotions. For instance, *guilt* and *pride* are both materialized in situations with high *self-agency*, with the former happening in *pleasant* situations while the latter comes with an *unpleasant* experience. These mappings, which align closely with prior findings in appraisal theory (Wondra and Ellsworth, 2015), are extracted from our dataset by averaging appraisal scores for each emotion label.

We begin by training linear probes for appraisal signals within the model representations. In contrast to the linear classification probes used in Section 5, here we solve multiple independent regression tasks for each appraisal. More formally, consider a set of n appraisals and let $\mathbf{v}_a \in \mathbb{R}^d$ represent the weight vector corresponding to the appraisal $a \in \{1, \cdots, n\}$. Let $\mathbf{x}$ be an activation vector, as introduced in Section 4. We train the regression weight $\mathbf{v}_a$ and the bias $b_a \in \mathbb{R}$ such that,

$$\hat{r}_a = \mathbf{v}_a^\top \mathbf{x} + b_a,$$

where $\hat{r}_a \in \mathbb{R}$ is the estimate for the appraisal score a, given the input $\mathbf{x}$. We train a separate appraisal probe for each appraisal at the hidden states of each layer. The weight matrices obtained through this process serve as representations of the corresponding appraisal, encoding features of the appraisal concept at the activations of the specified locations. The success of appraisal probing highly depends on whether the target concept is linearly detectable in the targeted activation. We provide the appraisal probing results in Appendix E, showing that the appraisal signals are not linearly detectable at earlier layers but are strongly present as we approach the hidden state of the final layers.